HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented wideband balun architectures using electromagnetic simulation tools, achieving target impedance matching across a **2-18 GHz** frequency range.
- Fabricated prototype balun structures using advanced semiconductor processing techniques and conducted RF/mm-wave characterisation, validating simulation models with experimental data.
- Investigated electromagnetic interactions and parasitic effects within advanced IC packaging substrates, identifying key coupling mechanisms that impact balun performance by up to **3 dB**.
- Developed and validated new simulation and measurement methodologies for RF components, reducing characterisation time for new designs by approximately **20%**.
- Optimised balun performance for package-level integration through iterative design refinement and simulation, improving insertion loss by **1.5 dB** at the upper frequency band.
- Collaborated with AMD researchers and engineers on joint research programmes, contributing to the translation of fundamental research into practical semiconductor applications.
- Managed a personal research project focused on wideband balun technologies and IC packaging integration, documenting findings and presenting progress to supervisors and industry partners.

SKILLS

RF & Microwave – Wideband Balun Design, RF/mm-wave Characterisation, Impedance Matching, Insertion Loss, Electromagnetic Simulation, Parasitic Effects, IC Packaging Integration

Semiconductor Fabrication – Device Fabrication, Prototype Validation, Semiconductor Processing, Nanotechnology, Microelectronics

Simulation & Modelling – Electromagnetic Modelling, RF Simulation, Design Validation, Measurement Methodologies

Programming & Scripting – Python, R, SQL

Data Analysis & Tools – Power BI, Tableau, Pandas, NumPy, Scikit-learn

General Software & Tools – Jira, Confluence, Git, Jenkins, Docker, Linux

Domain Knowledge – Integrated Circuits (ICs), High-Frequency Systems, Deep-Tech Research, Advanced Packaging

PROJECTS

Wideband Balun Design for Next-Gen ICs

- Investigated multiple balun architectures (e.g., Marchand, branch-line), simulating their performance limits in the **2-18 GHz** range using electromagnetic simulation software.
- Fabricated and tested prototype balun structures on custom substrates, achieving measured insertion loss below **2.5 dB** and return loss better than **10 dB** across the target bandwidth.
- Documented design trade-offs and experimental results, providing clear insights into the impact of packaging parasitics on balun performance for integration into advanced IC designs.

Advanced IC Packaging Integration Study

- Analyzed electromagnetic coupling and parasitic effects between balun components and surrounding IC package features, identifying critical integration challenges.
- Optimized balun layout and interconnects for integration into advanced packaging technologies, demonstrating a potential reduction in signal degradation by approximately **15%**.
- Developed a Python script to automate the analysis of S-parameter data from characterisation, improving the efficiency of performance evaluation by **30%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)